Minqi (Christelle) Xiong

Data Scientist | Clinical AI | Real-World Healthcare Data

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Google Scholar | LinkedIn | GitHub | Personal Website

Education

Johns Hopkins School of Medicine — **Baltimore, MD**
M.S. in Health Sciences Informatics – Research — *Aug. 2024 – Aug. 2026 (Expected)*
Certificate in Implementation Science and Research Practice

The Chinese University of Hong Kong — **Shenzhen, China**
B.Sc. in Data Science & Big Data Technology — *Sep. 2020 – May 2024*
Dean's List · First Prize, Netflix Recommendations Competition · First Prize, Kaggle COVID-19 Classification

Technical Skills

Programming	Python · R · SQL
ML Methods	Predictive modeling · survival analysis · causal inference · longitudinal modeling · natural language processing
ML Frameworks	PyTorch · Scikit-learn · XGBoost · LightGBM · spaCy
Healthcare Data	EHR · administrative claims · disease registries
Data Standards	OMOP CDM · FHIR
Clinical Terminologies	ICD-10 · RxNorm · CPT · LOINC
Data Infrastructure	SQL Server · Google BigQuery · Databricks · Docker · Git
Software	VS Code · MATLAB · TreeAge · Overleaf

Research Experience

Decision Sciences Lab, Johns Hopkins School of Medicine — **Baltimore, MD**
Graduate Researcher, PI: Dr. Harold P. Lehmann — *Jan. 2025 – Present*

Project *Master's Thesis — Conceptual and Computational Framework of Disease Severity*

- Conducting a PRISMA-ScR–guided scoping review to systematically synthesize how disease severity is defined and conceptualized across diverse clinical domains
- Developing a unified conceptual framework for representing multilevel and multidimensional disease severity to support computational modeling and clinical decision support
- Designing a temporal state-transition modeling framework to characterize dynamic disease severity trajectories across disease stages and patient contexts

Project *Open-source Systematic & Scoping Review Platform*

- Developing an open-source platform to support systematic and scoping reviews, integrating literature management, screening workflows, and reproducible evidence synthesis pipelines
- Designing ontology-informed structures for representing disease concepts and evidence relationships
- Implementing engineering components for automated screening workflows, corpus management, and reproducible research pipelines
- Project repository: Evidence Synthesis Platform

Center for Population Health Information Technology (CPHIT) — **Baltimore, MD**
Graduate Researcher, PI: Dr. Hadi Kharrazi — *Oct. 2024 – Dec. 2025*

Project *Racial Disparities in Post-Hysterectomy Emergency Department Visits*

- Processed Healthcare Cost and Utilization Project (HCUP) comprising 48M+ inpatient and outpatient records
- Designed cohort selection pipelines to identify about 200K hysterectomy cases for further analysis

- Developed multivariable logistic regression and sensitivity analysis workflows in R to evaluate racial disparities in the case of 30-day emergency department utilization
- Generated statistical results and analytical tables supporting a first-author peer-reviewed journal article published in *International Journal of Gynecology and Obstetrics*

Project *Diagnostic Concordance Between Administrative Claims and Electronic Health Records*

- Analyzed Optum Labs data covering over 17 million patients across 56 healthcare organizations
- Evaluated concordance and discrepancies between diagnosis codes recorded in claims and EHR systems
- Developed large-scale data processing and statistical analysis pipelines for healthcare data quality assessment

Myositis Center, Johns Hopkins Rheumatology **Baltimore, MD**
Graduate Researcher, PI: Dr. Christopher A. Mecoli *Oct. 2024 – Present*

Project *OHDSI Network Study of Dermatomyositis Adverse Events*

- Developed computable phenotypes and cohort discovery pipelines for dermatomyositis using the OMOP Common Data Model (CDM) and OHDSI ATLAS to enable multi-institutional observational research
- Contributing to study design and comparative effectiveness analyses evaluating immunosuppressant therapies and medication-related adverse events using large-scale OMOP-based observational data
- Assisting an R package to support cohort evaluation across multi-institutional OHDSI databases
- Project repository: Guideline-driven Evidence in Rheums Infxn Study

Project *Steroid Dose Calculation and Exposure Profiling*

- Led the development of a framework for estimating corticosteroid daily dosage from EHR databases
- Designed data pipelines to extract and normalize dosing information from medication prescriptions
- Implemented and evaluated NLP parsers and LLM-based approaches for dosage information processing
- Developing an open-source R package for reproducible steroid exposure calculation and multi-site evaluation across distributed OMOP partner databases
- Project repository: Steroid Dosage Calculation

Human Language Technology Laboratory **Shenzhen, China**
Research Assistant, PI: Dr. Zhiheng Wu *Sep. 2022 – Jul. 2023*

Project *Accent and Speech Voice Conversion*

- Collected an audio data set from 40 volunteers and pre-processed and refined the 120 hours audio dataset
- Co-developed SoftVQ, a vector quantization framework for non-parallel many-to-many voice conversion
- Built an end-to-end deep learning pipeline integrating BN-PNet, SoftVQ modules, and autoregressive decoding
- Achieved improvements of 76.3 points in perceived audio quality and 1.1 points in speaker similarity in ABX evaluations using JavaScript-based questionnaires

Innovation & Entrepreneurship

Igor AI **United States**
Co-Founder *2025 – Present*

- Led the early-stage product design for an LLM-based platform supporting AI-assisted project management
- Contributed to discussions on system architecture and use cases for generative AI in collaborative platforms
- Exploring applications of LLMs to streamline task coordination and communication between team members and management

Suzhou Zhiyan Yikang Medical Technology Co., Ltd. **China**
Co-Founder *2024 – Present*

- Co-founded a healthcare technology startup focused on developing diagnostic reagents for Alzheimer's disease
- Led the design and execution planning of clinical trials for diagnostic testing
- Led the development of a disease-specific Alzheimer's database to support clinical research and real-world data analysis

Hexcite: Excited for Healthcare, Johns Hopkins Technology Ventures **Baltimore, MD**
Technical Product Lead, GlioNet *Jan. 2025 – May 2025*

- Designed clinical UI and workflow integration for a precision medicine clinical decision support tool predicting glioblastoma survival and gene mutations from brain MRI and clinical features
- Contributed to product strategy, business model development, and product–market fit analysis
- Selected for **Audience Favorite Award** at the Hexcite Healthcare Innovation Program

EDGE Global AI & Web3 Summit, VertexLab **Hong Kong**
Conference Organizer & Interviewer *Apr. 2023 – Oct. 2023*

- Organized and moderated speaker panels for an international AI and Web3 technology summit
- Conducted live interviews with industry leaders on emerging AI technologies and startup ecosystems

Selected Publications

H-index: 4 † = *co-first authorship*

Peer-Reviewed Journal Articles

1.: **Xiong C**, Wang Y, Dun C, Borahay M, Kharrazi H. Assessing the impact of racial disparities on 30-day emergency department visits due to hysterectomy complications: A multi-state retrospective cohort study. *Int J Gynaecol Obstet*. 2026 (in-press). [**First & Corresponding Author**]

2.: **Xiong C**, Pittell H, Kitchen C, Lasser E, Kharrazi H. Synergy of diagnosis coding between administrative claims and electronic health records of large patient populations across multiple healthcare organizations. *JAMIA Open*. 2026 (in-press).

3.: Lee H, **Xiong C**, Baughman D, Dun C, Tong J, Martin B, Lehmann H, Nagy P. A multidimensional hierarchical framework for sources of bias in real-world healthcare evidence: a scoping review. *J Biomed Inform*. 2026 Feb;174:104989. doi:10.1016/j.jbi.2026.104989. Epub 2026 Jan 20. PMID: 41571173.

4.: Boyer T, Barrett R, **Xiong C**, Bhansali R, Wallace A, Fang M, Vaught AJ, Selvin E, Ndumele C, Blaha M, Michos E, Coresh J, Minhas A. Hypertensive Disorders of Pregnancy and Premature Cardiovascular Disease in a Diverse Cohort of Young US Women. *Circulation*. 2026 Feb 17;153(7):480-492. doi:10.1161/CIRCULATIONAHA.125.078057. Epub 2025 Nov 8. PMID: 41697979; PMCID: PMC12912777.

5.: Tao Y, **Xiong M**, Peng Y, Yao L, Zhu H, Zhou Q, Ouyang J. Machine learning-based identification and validation of immune-related biomarkers for early diagnosis and targeted therapy in diabetic retinopathy. *Gene*. 2025 Jan 20;934:149015. doi:10.1016/j.gene.2024.149015. Epub 2024 Oct 18. PMID: 39427825.

6.: Shi T, Ma J, Yu Z, Xu H, **Xiong M**, Xiao M, Li Y, Zhao H, Kong G. Large language models in critical care medicine: a scoping review. *JMIR Med Inform*. 2025 Nov 24;13:e76326. doi:10.2196/76326. PMID: 41284992; PMCID: PMC12778902.

7.: Chen D, Couperus CJ, Lee S, Barrett R, Lee H, **Xiong MC**, Wang Y, Wang Q, Lehmann HP. Evaluation of large language model performance in assessing health economic study quality. *J Health Econ Outcomes Res*. 2025 Oct 24;12(2):154-162. doi:10.36469/001c.145214. PMID: 41146963; PMCID:PMC12554303.

Manuscripts Under Review or In Preparation

8.: Dobbins N†, **Xiong C**†, Lan K, Yetisgen M. LLM-based agents for automated research reproducibility in Alzheimer's disease. *J Med Internet Res*. [**Co-first Author; Under Review**].

9.: **Xiong C**, Lehmann HP. Conceptualizing disease severity: a scoping review and proposed conceptual framework. *J Biomed Inform*. [**In Preparation**].

10.: **Xiong C**, Barrett R, Boyer T, Martin B, Minhas A. Evaluation of pregnancy episode identification algorithms in real-world healthcare data. *JAMIA*. [**In Preparation**].

11.: Castner MD, Kitchen C, **Xiong C**, Bittle M, Nestadt PS, Wilcox HC, Kharrazi H. Assessing the utility of social determinants of health data in suicide prediction models. *Arch Suicide Res*. [**Under Review**].

12.: Castner MD, Kitchen C, **Xiong C**, Bittle M, Nestadt PS, Wilcox HC, Kharrazi H. Assessing the utility of social determinants of health and access data in ecological suicide prediction models. *Soc Psychiatry Psychiatr Epidemiol*. [**Under Review**].

Conference Full Papers

15.: **Xiong M**, Wu Z. Soft Vector Quantization for Non-Parallel Any-to-Many Voice Conversion. *Speech and Language Processing Technology Conference*. 2023. [Manuscript] [Poster]

16.: Zhang H, **Xiong M**, Shi T, Liu W, Xu H, Zhao H, Kong G. Reinforcement Learning-based Decision-making for Renal Replacement Therapy in ICU-acquired AKI Patients. *ACM KDD 2024 Workshop on AI for Decision Support in Healthcare*. [Manuscript] [Poster]

Oral Presentations & Posters

O1.: **Xiong C**, Wang Y, Dun C, Kharrazi H. Racial disparities in 30-day ED visits due to hysterectomy complications. *AMIA Annual Symposium 2025*. **[Podium; Selected for Recording & On-Demand Dissemination]** [Abstract] [Slides]

O2.: **Xiong C**, Martin B, Westlund E, Kelly W, Mecoli C. AgentDose: Toward Accurate and Scalable Steroid Dose Extraction in OMOP Using NLP Parsers and LLM Agents. *OHDSI Global Symposium 2025*. **[Nominee, Best Community Contribution Showcase]** [Brief Report] [Poster]

O3.: Lee H, **Xiong C**, et al. A Framework for Understanding Bias in Real-World Clinical EHR Data. *AMIA Annual Symposium 2025*. [Poster; Elected DEI Communications Presentation] [Poster]

Teaching & Mentoring

Johns Hopkins School of Medicine
Graduate Teaching Assistant *2025–2026*
Courses supported:
- ME.250.782 Observational Health Research Methods on Medical Records (OMOP)
- ME.250.861–864 Health Sciences Informatics Research Methods I–IV
- ME.250.788 Observational Research Methods in R

Self-Employed
Independent Academic Advisor *2021–Present*
Advised and mentored 27 high school and undergraduate students on undergraduate and graduate school applications, including CV preparation, personal statements, program selection, and long-term academic career planning

Honors & Awards

2025: Selected Presentation, AMIA Annual Symposium (recorded presentation and on-demand dissemination)

2025: Best Community Contribution Showcase Nominee, OHDSI Global Symposium (AgentDose)

2025: Audience Favorite Award, Hexcite Healthcare Innovation Program, Johns Hopkins Technology Ventures

2023: Dean's List, The Chinese University of Hong Kong, Shenzhen

2023: First Prize, Netflix Recommendation Algorithm Competition

2023: First Prize, COVID-19 Behavioral Differences Classification Challenge (Kaggle)

2020: Bowen Scholarship (merit-based, $4,000/year)

Professional Service

- Reviewer for journals: *Medical Care*, *JAMIA Open*
- Reviewer for conferences: AMIA Annual Symposium, Amplify Informatics Conference, Clinical Informatics Conference, IEEE International Conference on Health Informatics (ICHI)
- Member of American Medical Informatics Association (AMIA), Observational Health Data Science and Informatics (OHDSI), and American Psychological Association (APA)
- Program committe member of IEEE-ICHI 2026